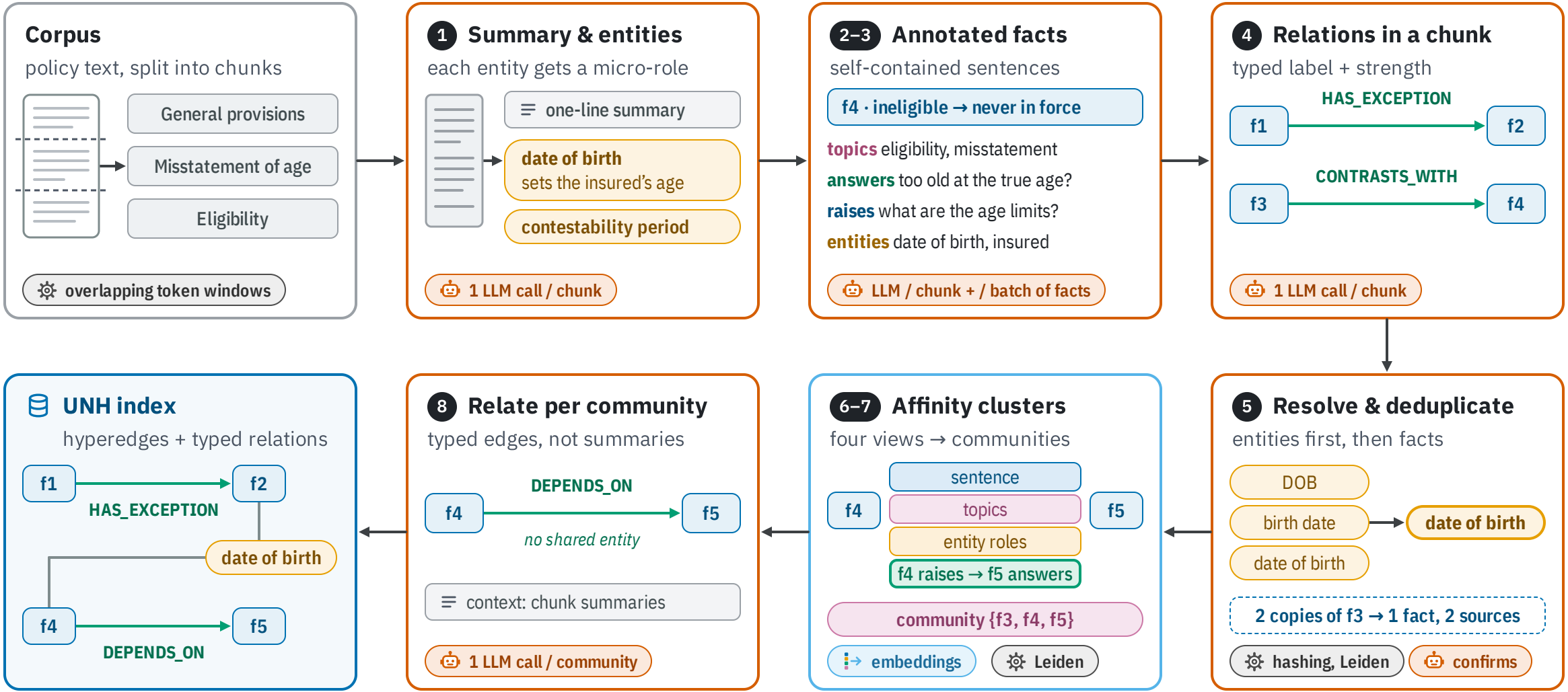

Corpus
policy text, split into chunks
General provisions
Misstatement of age
Eligibility
overlapping token windows
1 Summary & entities
each entity gets a micro-role
one-line summary
date of birth
sets the insured's age
contestability period
1 LLM call / chunk
2–3 Annotated facts
self-contained sentences
f4 · ineligible → never in force
topics eligibility, misstatement
answers too old at the true age?
raises what are the age limits?
entities date of birth, insured
LLM / chunk + / batch of facts
4 Relations in a chunk
typed label + strength
HAS_EXCEPTION
f1
f2
CONTRASTS_WITH
f3
f4
1 LLM call / chunk
5 Resolve & deduplicate
entities first, then facts
DOB
birth date
date of birth
date of birth
2 copies of f3 → 1 fact, 2 sources
hashing, Leiden
confirms
6–7 Affinity clusters
four views → communities
f4
sentence
topics
entity roles
f4 raises → f5 answers
f5
community {f3, f4, f5}
embeddings
Leiden
8 Relate per community
typed edges, not summaries
f4
DEPENDS_ON
f5
no shared entity
context: chunk summaries
1 LLM call / community
UNH index
hyperedges + typed relations
f1
HAS_EXCEPTION
f2
date of birth
f4
DEPENDS_ON
f5